

Week 11 Lab Assistant Preparation Guide

Topic: Legal, ethical, and social issues in technology. This week is discussion and report based, not mainly terminal based.

Main task: students read one assigned case, identify legal, ethical, and social issues, research evidence, discuss trade-offs, and write recommendations.

```
week11/  
|-- README.md  
|-- report.md  
`-- cases/  
    |-- CASE-1.md  
    |-- CASE-2.md  
    |-- CASE-3.md  
    `-- CASE-4.md
```

1. What Week 11 is about

- Students analyse real-world technology decisions beyond whether the system technically works.
- They consider who is affected, what could go wrong, whether the decision is legal, whether it is ethical, and what the company should do next.
- The four case topics are AI hiring discrimination, workplace location tracking, police access to smart home data, and autonomous warehouse robot safety.

Simple framing for students: You are acting like technology consultants. A company has a problem. Your job is to explain the risks and recommend practical next steps.

2. Relevant repo folders and files

README.md

- Explains the overall group task.
- Students work in groups of three.
- Suggested roles: Research Lead, Discussion Lead, and Scribe/Synthesiser.
- Students should rotate roles halfway through.
- Students complete report.md and may prepare a short 2-minute presentation.

report.md

This is the main template students fill in. It includes group roles, case summary, stakeholders, issues, evidence, analysis, recommendations, implementation notes, reflection, and references.

cases/CASE-1.md - AI hiring system

- Main issue: an AI recruitment tool may reproduce unfair patterns from historical hiring data.
- Older applicants and applicants from some universities are ranked lower.
- Key ideas: biased historical data, indirect discrimination, fairness, explainability, and human oversight.

cases/CASE-2.md - Location tracking at work

- Main issue: drivers are tracked using a smartphone app for safety, but workers feel constantly monitored.
- Key ideas: GDPR, data minimisation, transparency, employee monitoring, proportionality, and worker trust.

cases/CASE-3.md - Smart home device and police access

- Main issue: police request 48 hours of smart speaker audio logs after a burglary.
- Key ideas: privacy, consent, lawful requests, data minimisation, domestic surveillance, and trust.

cases/CASE-4.md - Autonomous warehouse robots

- Main issue: robots have malfunctioned, firmware updates are skipped, and management wants to maintain throughput.
- Key ideas: workplace safety, risk assessment, automation, duty of care, and safety versus efficiency.

3. What students are expected to do

- Read their assigned case carefully.
- Identify the main facts and the decision that needs to be made.
- List stakeholders: people or groups affected by the decision.
- Separate legal, ethical, and social issues.
- Find evidence from credible sources such as official guidance, law pages, regulator guidance, professional codes, academic articles, or credible news.
- Discuss trade-offs such as privacy versus safety or efficiency versus fairness.
- Write 4-6 practical recommendations.
- Complete report.md and add references.
- Optionally prepare a 2-minute presentation: case summary, key issues, recommended action.

4. What you should personally understand before the lab

Legal vs ethical vs social

Legal = what the law requires or forbids

Ethical = what is fair, responsible, or morally right

Social = how it affects people, trust, jobs, communities, or society

Example for AI hiring: legal means possible discrimination risk; ethical means unfair automated decisions; social means reduced trust in AI recruitment.

Stakeholders

A stakeholder is anyone who can be helped, harmed, or influenced by the technology. Examples: users, employees, customers, managers, regulators, developers, vulnerable groups, and wider society.

Evidence

Students should not rely only on opinions. Good evidence can come from GOV.UK, ICO, HSE, Equality and Human Rights Commission, ACM Code of Ethics, IEEE Code of Ethics, academic articles, and credible news.

Recommendations

A strong recommendation is specific and practical. Weak: "The company should be ethical." Strong: "Suspend automated rejection decisions until the model has been audited for age and university bias, then reintroduce it only with human review and regular fairness monitoring."

5. Common student mistakes and how to fix them

Only summarising the case: Ask: What is the problem? Who is harmed? What law or rule might apply? What should the company do?

Writing opinions without evidence: Ask: Which law, guidance, professional code, or credible source supports that point?

Confusing legal and ethical issues: Remind them: legal means rules/law; ethical means fairness, responsibility, and harm.

Forgetting stakeholders: Ask them to list primary stakeholders who are directly affected and secondary stakeholders who are indirectly affected.

Writing vague recommendations: Push them to write action + reason + expected benefit.

Ignoring trade-offs: Ask: who gets the benefit, who carries the risk, and what alternative would reduce harm?

Missing references: Tell them to add links or citations in the References section.

One student doing everything: Remind groups to rotate roles halfway through as the README says.

6. Concepts you should be ready to explain

Indirect discrimination: A rule or system looks neutral but disadvantages a protected group in practice.

Bias in historical data: If old company decisions were biased, an AI trained on those decisions can learn the same bias.

Data minimisation: Only collect the data you truly need for the stated purpose.

Transparency: People should understand what data is collected, why, who can see it, and what decisions are made from it.

Proportionality: The action should not be more intrusive or risky than necessary.

Accountability: Someone must be responsible for checking the system, explaining decisions, and fixing harm.

Human oversight: A person should review important automated decisions, especially when they affect jobs, privacy, safety, or rights.

Risk assessment: Identify what could go wrong, how serious it would be, how likely it is, and what controls are needed.

7. Physical or software setup to check

- Students can access the week11 folder.
- Students can open their assigned CASE-X.md file.
- Students can open and edit report.md.
- Groups know which case they have.
- Groups know their roles and when to rotate them.
- Internet works for research.
- Students know how to add references or links.
- Presentation requirement is clear.
- Submission format is clear: GitHub, Moodle, or another platform.

If Git submission is required, be ready to explain:

```
git status
git add report.md
git commit -m "Complete week 11 case analysis"
git push
```

8. Simple explanations for beginners

What is this lab asking?: You are advising a company about a technology problem and recommending responsible next steps.

Good case summary: Explain what happened, who is involved, and what decision needs to be made.

Legal issue: Something that could break or trigger legal rules, such as discrimination, privacy, safety, or employment law.

Ethical issue: Something about fairness, harm, honesty, responsibility, and respect for people.

Social issue: The wider effect on society, trust, jobs, public safety, or vulnerable groups.

Recommendation: A clear action the company should take, not just a complaint that the technology is bad.

9. Case-by-case guidance

Case 1 - AI Hiring System

- Likely issues: indirect discrimination, lack of fairness, biased historical data, weak explainability, and lack of human oversight.
- Good recommendations: pause automated rejection, audit outcomes by age and education background, retrain with better data, add human review, create appeal process, monitor fairness metrics.

Case 2 - Location Tracking

- Likely issues: GDPR transparency, data minimisation, weak consent, purpose creep, surveillance stress, and damaged worker trust.
- Good recommendations: rewrite monitoring notice, limit data to safety needs, separate safety monitoring from punishment, consult workers, set retention limits.

Case 3 - Smart Home Device

- Likely issues: privacy, sensitive family audio, lawful request scope, proportionality, and public trust in smart devices.
- Good recommendations: do not release all 48 hours automatically, request narrower scope, notify owner where allowed, disclose only relevant data, audit the disclosure.

Case 4 - Warehouse Robots

- Likely issues: workplace safety duties, skipped maintenance, repeated malfunctions, productivity pressure, and worker trust.
- Good recommendations: reduce robot operation in high-risk areas, install firmware updates, investigate malfunctions, update risk assessment, improve emergency stop procedures.

10. Short checklist for you before the lab

- ☐ Know the four cases
- ☐ Know where report.md is
- ☐ Prepare simple definitions of legal, ethical, social
- ☐ Prepare examples of stakeholders
- ☐ Prepare examples of strong vs weak recommendations
- ☐ Remind students to use evidence, not just opinions
- ☐ Remind students to rotate group roles
- ☐ Check whether presentation is required
- ☐ Check whether references/links are required
- ☐ Check whether submission is through GitHub, Moodle, or another platform

11. Missing or incomplete parts in the repo

- Presentation template is mentioned, but docs/presentation-template.md was not visible in the week11 folder. Students may need a simple 3-part presentation structure unless staff provide a template.
- Assigned cases are not specified in the repo. Staff need to assign cases or let groups choose.
- Source expectations are vague. A practical minimum is 2-4 credible sources per group, but confirm with staff.
- Report length is not fully specified. Encourage concise bullet points.
- Git submission is not confirmed. Check whether students submit through GitHub, Moodle, or another platform.

Main thing to remember

Week 11 is not about commands. It is about clear thinking around technology decisions. Keep students from writing shallow summaries. Push them to identify stakeholders, harms, benefits, evidence, trade-offs, and practical recommendations.